

CURRICULUM

3rd – 8th Semester July 2023 Admissions onwards

B.Tech. in DATA SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DR B R AMBEDKAR NATIONAL INSTITUTE OF TECHNOLOGY JALANDHAR

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VISION

To be recognized globally for imparting computer science education and research of high distinction, both of value and relevance to society.

MISSION

M1: To impart contemporary knowledge and skill relevant to the field of Computer Science and Engineering to maximize employability and potential.

M2: To strengthen multifaceted competence in the different core and allied areas of Computer Science in order to nurture creativity, innovations and out-of-the-box thinking.

M3: To promote research and expertise in Computer Science and Engineering in order to serve the needs of Industry, Government and Society and motivate the students for 5

M4: To inculcate ethical, human, and social values through co-curricular and extra-curricular activities for holistic nation building.

Program Educational Objectives

PEO-1: To create and sustain a community of learning in which students acquire knowledge and apply in their concerned fields with due consideration for ethical, ecological, and economic issues.

PEO-2: To provide knowledge based services so as to meet the ever-changing needs of industry and society at large.

PEO-3: To make the students understand, design and implement the concepts in multiple arenas.

PEO-4: To foster holistic growth of students that would provide ample E2 opportunities.

Program Outcomes

1. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem Analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3. **Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems.
5. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.
11. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
12. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Program Specific Outcomes

PSO-1: Analyse, design, develop, evaluate and apply mathematical basics, data structures and algorithms for modeling computer hardware/software to solve real-world and interdisciplinary research problems.

PSO-2: Demonstrate contemporary technologies, data analysis and computing skills for effective interpretation and decision making in the sustainable development of the society.

3rd Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0201	Data Structures and Algorithms	3	0	0	3	DC
2	CSDC0203	Object Oriented Programming	3	0	0	3	DC
3	CSDC0205	Computer Organization and Architecture	3	0	0	3	DC
4	CSDC0207	Data Visualization for Product Design	3	1	0	4	DC
5	CSDC0211	Mathematical Foundation for Data Science - I	3	1	0	4	DC
6	HMDC0201	Business Economics and Financial Accounting	3	0	0	3	DC
7	CSDC0231	Data Structures and Algorithms Laboratory	0	0	3	2	DC
8	CSDC0233	Object Oriented Programming Laboratory	0	0	2	1	DC
9	CSDC0235	Computer Organization and Architecture Laboratory	0	0	2	1	DC
10	CSDC0237	Data Visualization for Product Design Laboratory	0	0	2	1	DC
TOTAL			18	2	9	25	

4th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0202	Database Management Systems	3	0	0	3	DC
2	CSDC0204	Design and Analysis of Algorithms	3	0	0	3	DC
3	CSDC0206	Computer Networks	3	0	0	3	DC
4	CSDC0210	Natural Language Processing	3	0	0	3	DC
5	CSDC0212	Mathematical Foundation for Data Science - II	3	1	0	4	DC
6	PDFE0xxx	Foundation Elective	1	0	0	1	FE3
7	CSDC0232	Database Management Systems Laboratory	0	0	2	1	DC
8	CSDC0234	Design and Analysis of Algorithms Laboratory	0	0	3	2	DC
9	CSDC0236	Computer Networks Laboratory	0	0	2	1	DC
10	CSDC0240	Natural Language Processing Laboratory	0	0	2	1	DC
TOTAL			16	1	9	22	

5th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0301	Artificial Intelligence	3	0	0	3	DC
2	CSDC0305	Operating Systems	3	0	0	3	DC
3	CSDC0307	Data Mining and Analytics	3	0	0	3	DC
4	CSDC0309	Computer Vision	3	0	0	3	DC
5	CSDE03XX/ XXGE03XX	Department Elective – I/ Generic Elective - I	3	0	0	3	DE/GE
7	CSDC0335	Operating Systems Laboratory	0	0	2	1	DC
8	CSDC0337	Data Mining and Analytics Laboratory	0	0	2	1	DC
6	CSDC0339	Computer Vision Laboratory	0	0	2	1	DC
9	CSTR0300	Industrial Practical Training (after 4thSem)	0	0	0	2	TR
10	CSPR0301	Project-1 (Phase-I)	0	0	2	2	PR
TOTAL			15	0	8	22	

6th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0304	Information Security Systems	3	0	0	3	DC
2	CSDC0306	Machine Learning	3	0	0	3	DC
3	CSDC0308	Software Engineering	3	0	0	3	DC
4	CSDC0310	Deep Learning	3	0	0	3	DC
5	CSDE03XX/ XXGE03XX	Department Elective – 2/ Generic Elective - 2	3	0	0	3	DE/GE
6	CSDC0334	Information Security System Laboratory	0	0	2	1	DC
7	CSDC0336	Machine Learning Laboratory	0	0	2	1	DC
8	CSDC0338	Software Engineering Laboratory	0	0	2	1	DC
9	CSDC0340	Deep Learning Laboratory	0	0	2	1	DC
10	CSPR0302	Project-1 (Phase-II)	0	0	2	2	PR
TOTAL			15	0	10	21	

List of Department Electives for 3rd year

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1.	CSDE0302	Advanced DBMS	3	0	0	3	DE
2.	CSDE0306	Advanced Computer Networks	3	0	0	3	DE
3.	CSDE0308	Android Programming and Mobile Applications Development	3	0	0	3	DE
4.	CSDE0310	Software Project Management	3	0	0	3	DE
5.	CSDE0312	Network Security & Cyber Forensics	3	0	0	3	DE
6.	CSDE0316	Formal Language and Automata Theory	3	0	0	3	DE
7.	CSDE0318	Augmented Reality and Virtual Reality	3	0	0	3	DE

7th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDE04XX/ XXGE04XX	Department Elective – 3/ Generic Elective	3	0	0	3	DE/GE
2	CSDE04XX/ XXGE04XX	Department Elective – 4/ Generic Elective	3	0	0	3	DE/GE
3	CSDE04XX/ XXGE04XX	Department Elective – 5/ Generic Elective	3	0	0	3	DE/GE
4	CSTR0400	Industrial Practical Training (after 6th Sem)	0	0	0	3	TR
5	CSPR0401	Project-2	0	0	8	4	PR
TOTAL			9	0	8	16	

8th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDE04XX/ XXGE04XX	Department Elective – 6/ Generic Elective	3	0	0	3	DE/GE
2	CSDE04XX/ XXGE04XX	Department Elective – 7/ Generic Elective	3	0	0	3	DE/GE
3	CSDE04XX/ XXGE04XX	Department Elective – 8/ Generic Elective	3	0	0	3	DE/GE
4	CSPR0402	Project-2	0	0	8	4	PR
TOTAL			9	0	8	13	

List of Department Electives for 4th year

S. No	Specialization	Course Code	Course Title	Teaching Load			Credit
				L	T	P	
1.	Artificial Intelligence	CSDE0401/ CSDC0309	Computer Vision	3	0	0	3
2.		CSDE0402/ CSDC0310	Deep Learning	3	0	0	3
3.		CSDE0403	Reinforcement Learning	3	0	0	3
4.		CSDE0404	Modern AI tools	3	0	0	3
5.		CSDE0406	Human-Computer Interaction	3	0	0	3
6.	Data Analytics	CSDE0407	Big data analytics	3	0	0	3
7.		CSDE0408	Social Network Analysis	3	0	0	3
8.		CSDE0409	Optimization Techniques	3	0	0	3
9.	Software Engineering	CSDE0410	Software Quality Assurance and Automation Testing	3	0	0	3
10.		CSDE0411	Software Engineering Ethics and Tools	3	0	0	3
11.		CSDE0412	Green Computing	3	0	0	3
12.	Computer Networks	CSDE0413	Evolution of Advanced Wireless Networks	3	0	0	3
13.		CSDE0414	5G Networks: Architecture, Technologies, and Applications	3	0	0	3
14.		CSDE0415	Industry 4.0	3	0	0	3
15.	Information Security	CSDE0416	Blockchain Architecture & Use Cases	3	0	0	3
16.		CSDE0417	Ethical Hacking and Penetration Testing	3	0	0	3
17.	Soft Computing	CSDE0418	Soft Computing Techniques	3	0	0	3
18.		CSDE0419	Evolutionary Algorithms	3	0	0	3
19.	Internet of Things	CSDE0421	Internet of Things Concepts	3	0	0	3
20.		CSDE0422	Real-time systems	3	0	0	3
21.		CSDE0423	Embedded Systems	3	0	0	3
22.		CSDE0424	Intelligent Robotics	3	0	0	3
23.	Parallel and Distributed Computing	CSDE0425	Distributed Systems	3	0	0	3
24.		CSDE0426	Edge and Fog Computing	3	0	0	3
25.		CSDE0427	Quantum Computing	3	0	0	3
26.		CSDE0428	High-Performance Computing	3	0	0	3
27.	Other	CSDE0429	Digital Marketing	3	0	0	3
28.		CSDE0430	Computational Complexity	3	0	0	3
29.		CSDE0431	Drone Technology	3	0	0	3

3rd Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0201	Data Structures and Algorithms	3	0	0	3	DC
2	CSDC0203	Object Oriented Programming	3	0	0	3	DC
3	CSDC0205	Computer Organization and Architecture	3	0	0	3	DC
4	CSDC0207	Data Visualization for Product Design	3	1	0	4	DC
5	CSDC0211	Mathematical Foundation for Data Science - I	3	1	0	4	DC
6	HMDC0201	Business Economics and Financial Accounting	3	0	0	3	DC
7	CSDC0231	Data Structures and Algorithms Laboratory	0	0	3	2	DC
8	CSDC0233	Object Oriented Programming Laboratory	0	0	2	1	DC
9	CSDC0235	Computer Organization and Architecture Laboratory	0	0	2	1	DC
10	CSDC0237	Data Visualization for Product Design Laboratory	0	0	2	1	DC
TOTAL			18	2	9	25	

Course Contents**Unit 1: Introduction to Data Structures and Complexity Analysis**

Overview of data structures, algorithms, Asymptotic notations (Big O, Omega, Theta), Time and space complexity analysis, Best, average, and worst-case analysis, Time space trade-off, Abstract Data types.

Unit 2: Array and Linked List

Arrays, implementing basic algorithms on 1D array (insertion, deletion, searching, traversal), Implementing basic algorithms on 2D array (Addition, Subtraction, Multiplication, Transpose), Address calculation for multidimensional array, sparse matrix, array sorting: Bubble sort, insertion sort and selection sort, Linked List, Representation and Implementation of Singly Linked Lists, Two-way Header List, Traversing and Searching of Linked List, Overflow and Underflow, Insertion and deletion Algorithms, doubly linked list, Linked List in Array, Polynomial representation, and addition, Generalized linked list, Circular linked list, Garbage Collection and Compaction.

Unit 3: Stack and Queue

Stack, implementing stack using arrays and linked lists, implement basic operations on stack (create, push, pop, full, empty), Applications of stacks, expression evaluation (prefix and postfix), Conversion of Infix to prefix and Postfix Expressions, Queue, implementing queue using arrays and linked lists, implement basic operations on queue (create, insert, delete, full, empty), Priority Queue, Circular Queue.

Unit 4: Tree and Graph

Trees, Basic terminologies, Array and linked representation of binary tree, Binary tree traversal (in-order, pre-order, post-order), Threaded Binary Tree, Huffman algorithm, Binary search trees (BST), Insertion and Deletion in BST, Graphs, Graph terminologies and representation of graphs (adjacency matrix, adjacency list), Multigraph, Directed Graph, Traversals - Depth-First Search (DFS) and Breadth-First Search (BFS), Connected component, Minimum Spanning Tree (Prim's, Kruskal's), Shortest path algorithm, Topological sorting.

Unit 5: Hashing and Heap

Hash Table, Hash Functions, Collision Resolution Strategies, Hash Table Implementation, heap-based implementations (Min, Max), insertion and deletion in heap, and heap sort.

Course Outcomes

CO1: Understand the concepts of data structures, algorithm and analyse their time complexity.

CO2: Apply sequential data structures e.g. Array and Linked list to solve basic problems.

CO3: Apply and analyze stack and queue data structures to solve practical problems in real-life scenarios.

CO4: Apply variety of data structures, including trees, graphs, and hashing techniques, to address diverse and complex real-time computing problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	3	2	3						2		1	3	1
CO2	2	1	2							1		1	3	
CO3	3	2	3	3			2			2		3	3	1
CO4	3	2	3	3			2			2		3	3	1

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Cormen, Thomas H., et al. Introduction to Algorithms. 4th ed., The MIT Press, 2022.
2. Tenenbaum, Aaron M., Yedidyah Langsam, and Moshe J. Augenstein. Data Structures Using C. 1st ed., Pearson Education, 2019.
3. Forouzan, Behrouz A., and Richard F. Gilberg. C Programming and Data Structures. 3rd ed., Cengage Learning India Pvt. Ltd., 2022.
4. Horowitz, Ellis, and Sartaj Sahni. Fundamentals of Data Structures in C++. 2nd ed., Universities Press, 2008.
5. Lipschutz, Seymour. Data Structures. 2nd ed., McGraw Hill, 2014.

Course Contents**Unit 1: Fundamentals of Object Oriented Programming**

Object oriented thinking: Need for OOP Paradigm, Procedural programming vs object oriented programming, object oriented concepts. C++ stream classes: Unformatted IO operations, formatted IO operations, managing output with manipulators.

Unit 2: Functions and Classes

Functions, inline functions, reference variables, call by reference, Default arguments, function overloading, Class, Difference between C structure and class, specifying a class, Defining member functions: inside and outside class, scope resolution operator, Array within a class, an array of objects, Static data members and member functions, Object as function arguments, returning objects, Friend function, memory allocation for objects, pointer to members, pointer to object, this pointer local classes.

Unit 3: Constructors, Operator Overloading, and Inheritance

Constructor and destructor: Constructor, types of constructors: default, parameterized and copy constructor, constructor overloading, constructor with default parameter, dynamic initialization of objects, destructor, Operator overloading and Type Conversion: Defining operator overloading, overloading unary and binary operator, Data Conversion: Basic to User Defined, User defined to basic, Conversion from one user-defined to other. Inheritance and polymorphism: Base class, derived class, visibility modes, derivation and friendship, Types of inheritance, Containership, virtual function binding, pure virtual functions, Abstract class, pointer to derived class.

Unit 4: File Handling, Exception Handling, and Templates

Working with files: Classes for file stream operations, opening and closing files, File opening modes, file Pointers, Error handling during file operations, and command line arguments. Exception Handling: Introduction to Exception Handling, Handling Exceptions with try-catch Blocks, Catching Specific Exception Types, Rethrowing Exceptions, Standard Exceptions and Custom Exception Classes. Templates: Class template, class template with parameter, function template, function template with parameter. Advanced OOP Concepts: Overview of the Standard Template Library (STL), Containers: vector, list, set, multiset, map, multimap.

Course Outcomes

- CO1: Understand the significance of object-oriented programming and distinguish it from procedural programming.
- CO2: Utilize C++ stream classes for various input and output operations effectively.
- CO3: Analyze and implement functions and classes, including function prototyping and member functions within classes.
- CO4: Evaluate and implement constructors, destructors, and operator overloading techniques in C++ programming.
- CO5: Apply inheritance, polymorphism, and exception-handling concepts to solve complex programming problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 11	PSO 12
CO1	1		1	2	2						2			1
CO2	2	2	2		2				1		1		1	
CO3	2	2	3	2	2			1			1		3	1
CO4	2	2	2	2	3			1			1		2	2
CO5	2	2	3	3	1	2	2	1	1		2	2	3	2

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Balagurusamy, E. Programming in ANSI C++. 7th ed., Tata McGraw-Hill Education, 2019.
2. Lafore, Robert. Object-Oriented Programming in C++. 5th ed., Pearson Education, 2019.
3. Stroustrup, Bjarne. Programming: Principles and Practice Using C++. Pearson Education, 2014.
4. Lippman, Stanley B., Josée Lajoie, and Barbara E. Moo. C++ Primer. 5th ed., Addison-Wesley, 2013.
5. Meyers, Scott. Effective Modern C++: 42 Specific Ways to Improve Your Use of C++11 and C++14. O'Reilly Media, 2014.

Course Contents

Unit 1: Introduction

Historical background, Architectural and Technological development, Performance measures. Functional Units of a computer: CPU, Memory, Input-output subsystems, and Control Unit.

Unit 2: Central Processing Unit

Registers, Instruction set, instruction types, Instruction length, Instruction formats, overview of Addressing modes, Instruction cycle, Instruction pipelining, Pipelining hazards, RISC versus CISC, Instruction-level Parallelism and Superscalar processors.

Unit 3: Control Unit

Micro-operations, Control of the processor, Hardwired and Micro-programmed control approaches.

Unit 4: Memory Organization

Introduction to Memory hierarchy, Importance of temporal and spatial locality, Main memory, Cache organization and performance, Associative memory, Cache coherence, Virtual memory, Memory interleaving, Implementing cache controllers, RAID.

Unit 5: Input/Output

External devices, Input/Output function, Interrupts, Programmed Input/Output, Interrupt-Driven Input/Output.

Course Outcomes

CO1: Understand the fundamentals of computer organization and its relevance to classical and modern problems of computer design.

CO2: Understand the architecture and functionality of the central processing unit.

CO3: Analyze various caching and memory system architectures.

CO4: To understand how I/O devices communicate with the Processing Unit.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	1											2	1
CO2	3	1											2	1
CO3	3	2	2	2									2	1
CO4	3	2	2	2									2	1

1 – Low, 2 – Medium, 3 - High

Recommended Books

1. Hennessy, John L., and David A. Patterson. Computer Architecture: A Quantitative Approach. 7th ed., Morgan Kaufmann, 2025.
2. Stallings, William. Computer Organization and Architecture. 11th ed., Pearson Education, 2022.
3. Hamacher, V. Carl, Zvonko G. Vranesic, Safwat G. Zaky, and Naraig Manjikian. Computer Organization. 6th ed., McGraw-Hill Education, 2012.
4. Shiva, Sajjan G. Computer Organization, Design, and Architecture. 5th ed., CRC Press, 2014.
5. Sarangi, S. R. Computer Organisation and Architecture. McGraw Hill India, 2014.
6. Mano, M. Morris. Computer System Architecture. 3rd ed., Pearson Education, 2017.

Course Contents**Unit 1: Introduction**

Data Types: Nominal, Binary, Ordinal, Numeric, Discrete, and Continuous, Value for visualization, Categorical, time series, and statistical data graphics, Introduction to Visualization Tools, Statistics of Data, Similarity & Distance Measures.

Unit 2: Data Gathering & Data Pre-processing

Data formats, various sources and types of data- text, video, audio, biology etc., parsing and transformation, Scalability and real-time issues, Data Quality, Consistency checking, Heterogeneous and missing data, Data Transformation and segmentation, Perspective transform, window transform, cleaning data, noise elimination, Data Integration, Feature selection and Dimensionality reduction, Normalization.

Unit 3: Statistics of Data

Descriptive Statistics, Measures of central tendency- mean, median, and mode; Measures of dispersion: variance, standard deviation, and range: Concept of Percentiles and quartiles.

Unit 4: Prescriptive Analytics

Creating data for analytics through designed experiments, Regression, Principal component analysis, Canonical correlation analysis, Analysis of variance, Creating data for analytics through active learning.

Unit 5: Data Visualization

Design Principles, Data-Ink, Visualization Typology, Geo Visualization, Data, Visualization Design, Tufte's design principles, Gestalt's law, Data Analysis and Insights finding, Critiques in data visualization, Data visualization with ethics. Visual Storytelling, Use of Data Visualization, Data Examples, Basic Charts, table, graph, histogram, pie-chart, area-plot, boxplot, scatter plot, bubble-plot, waffle charts, word clouds, Multidimensional Visualization, and Specialized visualization: Online Social Networks, Social Data Analysis, Geographical Data, Design of a dashboard.

Course Outcomes

- CO1: Understand the key techniques and theory used in visualization, including data models, graphical perception, and techniques for visual encoding and interaction.
- CO2: Analyse several common data domains and corresponding analysis tasks, including multivariate data, networks, text, and cartography.
- CO3: Apply the gained knowledge to build and evaluate visualization systems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	2												
CO2	3	3	1	1							1	1		
CO3	3	3	2	2							3	2	1	1

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Tufte, Edward R. The Visual Display of Quantitative Information. 2nd ed., Graphics Press, 2001.
2. Healy, Kieran. Data Visualization: A Practical Introduction. Princeton University Press, 2019.
3. Wilke, Claus O. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures. O'Reilly Publications, 2019.
4. Knaflic, Cole Nussbaumer. Storytelling with Data: A Data Visualization Guide for Business Professionals. John Wiley & Sons, 2015.
5. Han, Jiawei, Jian Pei, and Hanghang Tong. Data Mining: Concepts and Techniques. Morgan Kaufmann, 2022.

Course Contents

Unit 1: Linear Algebra for Data Science: Vectors and matrices Matrix operations: addition, subtraction, multiplication Vector spaces, Linear Transformations and subspaces, Kernel and range of a linear transformation Eigen values and eigenvectors, Matrix factorizations; Singular Value Decomposition (SVD) Applications in dimensionality reduction and feature engineering, Concept of feature reduction, Eigenvectors and eigenvalues in PCA, Application of PCA in data preprocessing.

Unit 2: Optimization Foundation: Mathematical optimization, types of optimizations – discrete and continuous optimization, least squares & linear programming, non-linear optimization, function optimization, optimization and machine learning, how to choose an optimization algorithm.

Unit 3: Convex sets: Affine and convex sets, operations that preserve convexity, convex function and basic properties, convex optimization, linear optimization, geometric programming, convex and non-convex optimization problems in machine learning/data science.

Unit 4: Optimization: local optimization vs. global optimization, premature convergence, creating visualization for function optimization, stochastic optimization algorithms, random search and grid search, Introduction to stochastic gradient descent (SGD), Understanding gradients and their role in optimization, Types of gradient descent: batch, stochastic, mini-batch, Convergence criteria and learning rate selection.

Unit 5: Metaheuristic Algorithms: Introduction to heuristic and metaheuristic optimization methods, comparison of different heuristic algorithms and its importance, Genetic algorithms, Differential evolution, Simulated annealing, Particle swarm optimization.

Course Outcomes

CO1: Develop a strong understanding of mathematical notation, terminology, and concepts relevant to data science.

CO2: Understand the principles and applications of linear algebra and matrix operations in data analysis and machine learning.

CO3: Analyze and apply optimization techniques for solving mathematical models and algorithms used in data science.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	2				2							2	1	1
CO2	1	2		2	2	2			2			2	2	3
CO3	1	3		2		1						2		3

Recommended Books

1. Lay, David C., Steven R. Lay, and Judi J. McDonald *Linear Algebra and Its Applications* (6th Edition). Pearson, 2024. ISBN: 9780135851258
2. Boyd, Stephen P., and Lieven Vandenberghe *Convex Optimization*. Cambridge University Press, 2004. (*Latest Edition*) ISBN: 9780521833783
3. Nocedal, Jorge, and Stephen J. Wright *Numerical Optimization* (2nd Edition). Springer, 2006. ISBN: 9780387303031
4. Rosen, Kenneth H. *Discrete Mathematics and Its Applications* (8th Edition). McGraw-Hill Education, 2018. ISBN: 9781259676512
5. Provost, Foster, and Tom Fawcett *Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking* (1st Edition). O'Reilly Media, 2013. ISBN: 9781449361327

Course Contents**Unit 1: Introduction to Business Economics**

The Economic Perspective, Microeconomics, Macroeconomics, Nature and Scope of Business Economics, Economic Optimization, Production Possibilities and Model, Unemployment, Growth, and the Future, Markets, Demand, Supply, Market Equilibrium, Changes in Supply, Demand and Equilibrium, Applications, Market Failures caused by Externalities and Asymmetric Information.

Unit 2: Consumer Behavior

Price Elasticity of Demand, Total Revenue Test, Determinants of Price Elasticity of Demand, Price Elasticity of Supply, Cross Elasticity and Income Elasticity of Demand, Law of Diminishing Marginal Utility, Theory of Consumer Behavior, Utility Maximization and the Demand Curve, Income and Substitution Effects, Applications.

Unit 3: Introduction to Financial Accounting

Introduction to Financial Accounting Meaning of Accountancy, Book Keeping and Accounting, Accounting Process, Objectives for Accounting, Differences between Book-keeping and Accounting, Users of Accounting Information, Limitations of Accounting, Accounting Principles- Accounting Concepts and Accounting Conventions, Accounting Standards - Scope and functions of Accounting Standards Board, International Financial Reporting Standards (IFRS).

Unit 4: Double Entry System of Book Keeping and Final Accounts

Types of Accounts, Accounting rules, Accounting Equation, preparation of journal, preparation of ledger accounts, subsidiary book and trial balance, Preparation of profit and loss account, balance sheet of sole proprietor and company (with and without adjustments), Adjustment Entries- closing stock, prepaid expenses, outstanding expenses, interest on capital and other adjustments, Cash flow statements.

Unit 5: Analysing Financial Statements

A Managerial Perspective- Understanding and analysing three basic financial statements: Balance Sheet, Income statement and Cash Flows, Horizontal and Vertical analysis, Ratio Analysis, Business Case.

Course Outcomes

- CO1: Apply the concepts of business economics in decision-making.
- CO2: Understand basic accounting concepts, such as double-entry bookkeeping, the accounting equation, debits and credits, and accrual accounting.
- CO3: Create the financial statements.
- CO4: Analyze accounting statements to evaluate, compare, and contrast a company's financial position, performance, and cash flow.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1						3							2	2
CO2		3												
CO3										3			2	
CO4		3		3		2							3	2

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Hirschey, Mark. Managerial Economics. 16th ed., Cengage Learning India, 2022.
2. McConnell, Campbell R., Stanley L. Brue, Sean M. Flynn, and B. R. Chaudhuri. Economics. 22nd ed., McGraw Hill Education (India), 2021.
3. Arora, R. K. Financial Accounting: Fundamentals, Analysis and Reporting. 2nd ed., Wiley India Pvt. Ltd., 2018.
4. Ramachandran, N., and R. K. Kakani. Financial Accounting for Management. McGraw Hill Education.
5. Narayanswamy, R. Financial Accounting: A Managerial Perspective. 6th ed., PHI Learning, 2017.

List of Experiments

1. Write a program to implement basic operations on 1D array e.g. Insertion, Deletion, Traversal, and Searching. Write a program to rearrange the array so that all negative numbers come before positive numbers.
2. Write a program to implement basic operations on 2D array/ Matrix e.g. Addition, Subtraction, Multiplication, Transpose. Write a program to calculate inverse of matrix.
3. Write a program using dynamic variable & pointers to implement insertion and deletion operation on a singly linked list consisting of the following information in each node. Student id (integer), student name (character string) & semester (integer). Write a program to search the data in above list using given student id.
4. Write a program to implement basic operations on doubly Linked List e.g. Insertion, Deletion, Traversal, and Searching. Write a program to rotate a single linked list by n positions.
5. Write a program to implement push, pop, and display operations on stack of integers. The program should print appropriate messages for stack overflow, stack Underflow & stack empty.
6. Write a c program to evaluate a valid postfix expression using a Stack, assume that the suffix /postfix expression is read as a single line consisting of non-negative single digit operands & binary arithmetic operands. The arithmetic operators are + (ADD), - (subtract), *(multiply) & / (divide).
7. Implementation of queue data structure using array and linked list with the basic functions of Create(), IsEmpty(), Insert(), Delete() and IsFull().
8. Implementation of stack operations to convert a given infix expression into its equivalent postfix expression. Implement the stack using an array. Write a program to implement basic operations of stack using queue data structure.
9. Write a program to construct a binary search tree of integers and traverse the tree using all the methods i.e. inorder, preorder & Postorder to display the elements in the tree.
10. Write a program to implement Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms.
11. Implementation of Kruskal and Prims Minimal Spanning Tree algorithms.
12. Write a modified version of DFS algorithm to print all the connected components of a graph.
13. Implement various advanced data structures such as min/max heaps and Hash tables.
14. Write a program to build a hash and map the values by incorporating collision resolution techniques.

****Mini Project may also be planned & carried out throughout the semester to understand the important various concepts of Data Structures and Algorithms.***

**This is only the suggested list of Practical's. Instructor may frame additional Practical's relevant to the course contents*

Course Outcomes

CO1: Apply sequential data structures e.g. Array and Linked list to solve basic problems.

CO2: Apply and analyse stack and queue data structures to solve practical problems in real-life scenarios.

CO3: Apply variety of data structures, including trees, graphs, and hashing techniques, to address diverse and complex real-time computing problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	1	2							1		1	3	
CO2	3	2	3	3			2			2		3	3	1
CO3	3	2	3	3			2			2		3	3	1

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Write programs to demonstrate array manipulation, function usage, and structure implementation.
2. Experiment with manipulators, reference variables, and call-by-reference in C++ programs.
3. Write programs to define, instantiate, and utilize classes and objects in C++.
4. Develop programs to showcase default, parameterized, and copy constructors, along with destructors.
5. Write programs to demonstrate friend functions, friend classes, and the concept of an empty class.
6. Implement unary operator overloading using both member and friend functions for user-defined classes.
7. Implement binary operator overloading using both member and friend functions for user-defined classes.
8. Develop programs to demonstrate the conversion from basic to custom data types and vice versa.
9. Implement single inheritance to create derived classes that inherit attributes and behaviors from a single base class.
10. Implement multiple inheritance and its complexities, including the use of virtual inheritance to resolve ambiguities.
11. Hands-on exercises and coding sessions to implement inheritance relationships and utilize derived classes in practical programs.
12. Implementing virtual functions in base and derived classes to demonstrate polymorphic behavior.
13. Explore try-catch blocks, throw statements, and exception specifications for robust error management. Writing code to handle specific exceptions and ensure program stability.
14. Write programs to perform basic file operations in C++, including reading from files, writing to files, appending data to files, and ensuring proper file closure, enabling effective data handling and manipulation in real-world applications.
15. Write programs to create generic classes and functions using templates, and practical sessions covering STL containers, algorithms, and iterators.

****Mini Project may also be planned & carried out throughout the semester to understand the important various concepts of Object Oriented Programming.***

**This is only the suggested list of Practical. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

- CO1: Understand basic and advanced C++ concepts, including efficient array, function, and structure manipulation, alongside advanced language features such as manipulators, reference variables, and call-by-reference, for effective programming.
- CO2: Define, instantiate, and utilize classes and objects, including constructors and destructors.
- CO3: Apply advanced C++ techniques, such as operator overloading, inheritance, and exception handling.
- CO4: Perform basic file operations in C++, ensuring proper data handling and manipulation.
- CO5: Utilize templates and STL containers, algorithms, and iterators for generic programming tasks.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	3			3			1					3	1
CO2		2	3		2				3	2	3		1	
CO3	2	1	3										2	
CO4	2	2		3	2			2					3	
CO5	2	2			3									2

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Verify various laws of Boolean Algebra. Implementation of a given Boolean function using logic gates in both SOP and POS forms.
2. Realize NAND and NOR Gates as universal gates.
3. Implementation of Adders, Subtractors, Multiplexer, and Demultiplexer.
4. Construct different types of Flip-flops and verify their truth tables.
5. Design and verify 4-bit BCD adder.
6. Design and verify 2-bit magnitude comparator.
7. Implementation and verification of Decoder, and Encoder.
8. Demonstrate the working of a BCD-to-7 Segment decoder and to verify the truth table.
9. Design and implement shift registers.
10. Design and implement synchronous and asynchronous counters.
11. To study the operation of Arithmetic and logic Unit using 74181 IC.
12. Implement cache memory approaches in ParaCache Simulator.

** This is only the suggested list of Practical. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

CO1: Realize the Boolean function using basic gates in both SOP/POS form.

CO2: Realize different combinational circuits with basic gates.

CO3: Analyse and design sequential digital circuits like flip-flops, registers, counters.

CO4: Explore the architecture and functionality of memory hierarchy.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	2	2	2	3								3	2
CO2	3	3	2	2	3								3	2
CO3	3	3	2	2	3								3	2
CO4	3	2	2	2	2								3	2

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Introduction to data visualization using Excel, Python and R.
2. Installation and hands-on exploration of Data Analysis and visualization tools: Power BI, Knime and Tableau.
3. Import a dataset from any repository and perform data cleaning. To Note: Add steps in procedure explaining how a dataset has been imported, followed by the steps for data cleaning.
4. Implement Exploratory Data Analysis techniques on a publicly available dataset.
5. Explore visualization tools for implementing basic plots: Box plot, histogram, heatmap, Scatter plot.
6. Perform feature engineering: Normalization, Binning. Should contain brief explanations about each function and their respective inference from the figures.
7. For features of a dataset, analyze the following values along with their inference:
 - a) correlation (chi-squared test and Pearson's correlation coefficient)
 - b) covariance.
8. Demonstrate the following:
 - a) Analyze time series data using any data visualization tool.
 - b) Generate a tree map in the tool for any dataset and interpret it.
9. Import required libraries on any data visualization tool required for clustering and classification.
10. Explain step wise how these libraries can be utilized for various ML algorithms.
11. Generate a data visualization dashboard

**This is only the suggested list of Practical. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

- CO1: Apply the key techniques and theory used in visualization, including data models, graphical perception, and techniques for visual encoding and interaction.
- CO2: Analyse several common data domains and corresponding analysis tasks, including multivariate data, networks, text, and cartography.
- CO3: Apply the gained knowledge to build and evaluate visualization systems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2
CO1	2	2												
CO2	3	2	1								1	1	1	1
CO3	3	2	1	2							2	2	2	1

1 – Low, 2 – Medium, 3 – High

4th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0202	Database Management Systems	3	0	0	3	DC
2	CSDC0204	Design and Analysis of Algorithms	3	0	0	3	DC
3	CSDC0206	Computer Networks	3	0	0	3	DC
4	CSDC0210	Natural Language Processing	3	0	0	3	DC
5	CSDC0212	Mathematical Foundation for Data Science - II	3	1	0	4	DC
6	PDFE0xxx	Foundation Elective	1	0	0	1	FE3
7	CSDC0232	Database Management Systems Laboratory	0	0	2	1	DC
8	CSDC0234	Design and Analysis of Algorithms Laboratory	0	0	3	2	DC
9	CSDC0236	Computer Networks Laboratory	0	0	2	1	DC
10	CSDC0240	Natural Language Processing Laboratory	0	0	2	1	DC
TOTAL			16	1	9	22	

Course Contents**Unit 1: Introduction**

An overview of the database management system, Database system Vs file system, Database system concepts and architecture, Data models, Schema and instances, Data independence, Overall Database Structure, ER model concepts, Notation for ER diagram, Mapping Constraints, Keys, Extended ER model: Generalization, Specialization and Aggregation, Reduction of ER diagram to tables.

Unit 2: Relational Data Model and Language

Relational data model concepts, Integrity constraints: Domain constraints, Entity integrity, Referential integrity, Keys constraints: Concepts of Super, Candidate, Primary and Foreign Keys, Relational algebra, Relational calculus: Tuple and Domain calculus, Introduction to SQL: SQL data types and literals. Types of SQL commands. Data Definition Language(DDL), Data Manipulation Language DML SQL operators, Queries, and subqueries. Aggregate functions. Insert, update, and delete operations. Joins, Unions, Intersection, Minus, Cursors, Triggers, Views, and Indexes.

Unit 3: Database Design & Normalization

Functional dependencies, normal forms, first, second, third normal forms, BCNF, Lossless join decompositions, Normalization using FD, MVD, and JDs, Query Processing and Optimization: Translating SQL Queries into Relational Algebra Operators, Algorithms for SELECT, PROJECT, JOIN and Set operations, Query Trees and Heuristics for Query Optimization.

Unit 4: Transaction Processing Concepts

Transaction processing, Transaction, and system concepts, Desirable properties of Transactions, Serializability of schedules: conflict & view the serializable schedule, Schedules based on recoverability, Database Recovery: Failure classification, Recovery from transaction failures, Log-based recovery: Deferred update and immediate update, shadow paging, Concurrency Control Techniques: Concurrency control, Two-phase locking Techniques for concurrency control, Time stamping protocols for concurrency control, Multi-version Concurrency Control, Validation protocol, and Multiple granularity schemes, Database Security: Database Security Issues, Discretionary Access Control, Mandatory Access Control, Role-Based Access Control, Encryption, and Public Key Infrastructures, Challenges to maintain Database Security.

Course Outcomes

CO1: Learn data models, conceptualize, and depict a database system using ER diagram

CO2: Understand and use data manipulation language to query, update, and manage a database

CO3: Apply normalization on the database design to eliminate anomalies

CO4: Do the analysis of database transactions and control them by applying ACID properties

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	3	1	2	2	1							2	3
CO2			3	3	2									
CO3		3	2	2	2			2	2					3
CO4		2		1	3			3	2			2		2

1 – Low, 2 – Medium, 3 - High

Recommended Books

1. Elmasri, Ramez, and Shamkant B. Navathe. Fundamentals of Database Systems. 7th ed., Pearson Education, 2016.
2. Korth, Henry F., Abraham Silberschatz, and S. Sudarshan. Database System Concepts. 7th ed., McGraw Hill, 2021.
3. Date, C. J. An Introduction to Database Systems. 8th ed., Addison Wesley, 2020.
4. Desai, Bipin C. An Introduction to Database Systems. Galgotia Publication, 2015.
5. Ramakrishnan, Raghu, and Johannes Gehrke. Database Management Systems. 3rd ed., McGraw-Hill, 2002.

Course Contents**Unit 1: Foundation of Algorithms**

Introduction to Algorithms and Recurrence Relation Analysis: Definition and characteristics of algorithms Importance of algorithm analysis, Amortized analysis Solving recurrence relations, Master's Theorem and substitution method, recursion-tree method, Advanced Sorting and order statistics: Merge sort, quick sort, sorting in linear time - Counting sort, Radix sort, Bucket sort, medians and order statistics.

Unit 2: Algorithm Paradigms

Divide and Conquer Algorithms: Recursive algorithms: Binary search; merge sort; quick sort, Closest pair problem, Strassen's matrix multiplication algorithm, Greedy Algorithms: Definition and characteristics of greedy algorithms Examples: fractional Knapsack problem, Huffman coding, Job scheduling, Dijkstra's algorithm, optimal merge pattern, Articulation point, Dynamic Programming: Principle of optimality Memorization and tabulation techniques Examples: 0-1 knapsack problem, longest common subsequence, matrix chain multiplication algorithm, Floyd Warshall algorithm, convex hull.

Unit 3: Backtracking and Branch and Bound

Backtracking: Overall technique, generation of combinatorial objects such as graphs, sets, permutations, graph colorings, cliques, Hamiltonian cycles, N Queen, Branch and Bound method: Overall method, the 0/1 knapsack problem, the job assignment problem, the traveling salesman problem, etc.

Unit 4: Advanced Topics

Advanced Data Structures: Red-Black trees, Binomial Heaps, Fibonacci Heaps, bi-connected component and max flow, Complexity Theory: P vs NP problem NP-completeness and NP-hardness Reduction techniques Cook's theorem

Course Outcomes

- CO1: Understand the theoretical foundations of algorithm design and evaluate various techniques for analyzing time and space complexity.
- CO2: Analyse and apply various algorithm design paradigms and problem-solving techniques.
- CO3: Analyse the efficiency and correctness of algorithms using mathematical methods and empirical analysis.
- CO4: Develop proficiency in implementing and evaluating algorithms for solving real-world problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	3	3	1	1							3	3	2
CO2	2	3	3	3	2						1	2	2	3
CO3	2	2	2	2	2				1		3	2	2	3
CO4	2	3	3	3	2				1		3	3	3	3

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Cormen, Thomas H., Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. Introduction to Algorithms. 4th ed., The MIT Press, 2022.
2. Kleinberg, Jon, and Éva Tardos. Algorithm Design. 1st ed., Pearson/Addison-Wesley, 2022.
3. Sedgewick, Robert, and Kevin Wayne. Algorithms. 5th ed., Addison-Wesley, 2021.
4. Levitin, Anany. The Design and Analysis of Algorithms. 3rd ed., Pearson, 2011.
5. Levitin, Anany, and Maria Levitin. Introduction to the Design and Analysis of Algorithms. 3rd ed., Pearson, 2011.

Course Contents**Unit 1: Introduction to Computer Networks**

Overview of computer network and its topologies, Network models: layered tasks, OSI model and its layers, TCP/IP protocol suite.

Unit 2: Physical and Data Link Layer

Physical Layer: Data and signals, Digital transmission, Analog transmission, Bandwidth utilization, Transmission media, Switching, Data Link Layer: Error detection and correction: block coding, cyclic codes, checksum; Data link control: framing, flow and error control, High level data link control, Point to point protocol, multiple access protocols: random access protocol, controlled access protocol, channelization protocol.

Unit 3: Network Layer

Network Layer: Network layer functions and services, Packet forwarding and routing, Routing algorithms: distance vector routing and link state routing and their implementations, Shortest path computation, Network layer protocols: IP & ICMP, IP addressing, Sub-netting and super netting (CIDR), Address mapping- ARP, RARP, BOOTP, and DHCP.

Unit 4: Transport Layer

Transport Layer: Transport Layer Services, User datagram protocol (UDP), Transmission control protocol (TCP), TCP connection management, Flow control, Congestion control, TCP congestion avoidance algorithms, Quality of service (QoS).

Unit 5: Session, Presentation and Application Layer

Session Layer: Session establishment, management, and termination, Dialog control, Synchronization, and Session Protocols, Presentation Layer: Data formatting and conversion, Character encoding schemes, Data compression, Encryption and decryption, Application Layer: Naming and addressing schemes, Domain name system, Hypertext transfer protocol, Remote login protocols- Secure shell and Telnet, Electronic mail: SMTP, POP3, File transfer protocol- Secure File Transfer: SFTP and FTPS, World wide web, Simple network management protocol.

Course Outcomes

- CO1: Understand the principles of computer networks, including network topologies, layering models, and protocol functions.
- CO2: Explain the functions and services of the network layer, including IP addressing, subnetting, CIDR, and routing protocols.
- CO3: Analyse IP addressing schemes and their role in efficient network design and optimization.
- CO4: Evaluate TCP & UDP transport layer protocols for connection management, flow control, congestion control, and reliability.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	2	2		2					2			1	
CO2	3	1	2	2	1					2			2	2
CO3	2	3	3	1	2					1			3	2
CO4	3	3	1	2	2			1		2			2	2

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Forouzan, Behrouz A. Data Communications and Networking. 6th ed., McGraw Hill, 2022.
2. Stallings, William. Data and Computer Communications. 10th ed., Pearson Education, 2021.
3. Forouzan, Behrouz A. TCP/IP Protocol Suite. 4th ed., McGraw-Hill Higher Education, 2009.
4. Tanenbaum, Andrew S. Computer Networks. 6th ed., Pearson Education India, 2022.
5. Kurose, James F., and Keith W. Ross. Computer Networking: A Top-Down Approach. 9th ed., Addison-Wesley, 2025.

Course Contents**Unit 1: Introduction - Different levels of NLP**

Text Normalization: Basic pre-processing, Word and sentence segmentation, Lemmatization, Stemming, Morphology.

Unit 2: Language Models

Language Models: n-gram models, smoothing techniques; Sequence Learning Tasks and Models. Examples of NLP tasks.

Unit 3: Statistical Model

HMM, MEMM, CRF; Syntactic Parsing: Regular and Context-Free Languages, Context-Free Parsing, CKY Algorithm; Dependency.

Unit 4: Parsing

Dependency Grammar, Graph-based dependency parsing, Transition-based dependency parsing; Vector-space Models: Word and Meanings, Distributional and Distributed Semantics, Lexical Semantic Analysis, GloVe, word2vec; Reference Resolution.

Unit 5: Applications

Text Classification and Categorization, Machine Translation, Information Retrieval, Question Answering Systems, Text Summarization, Text Generation, Sentiment Analysis, Speech Recognition, Language Understanding and Dialogue Systems.

Course Outcomes

- CO1: Understand fundamental concepts in natural language processing, including syntax, semantics, pragmatics, and discourse analysis.
- CO2: Demonstrate an ability to preprocess text data using techniques such as tokenization, stemming, lemmatization, stop word removal, and part-of-speech tagging.
- CO3: Develop and evaluate language models including statistical n-gram models, neural language models (e.g., LSTM, Transformer), and probabilistic context-free grammars.
- CO4: Use standard NLP tools and methods for text classification tasks, including sentiment analysis, topic classification, spam detection, and document categorization.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2			2					2			3	
CO2	3	3	2		3					2			3	2
CO3		2				3	3	2		2				3
CO4	2	3	3	3	3				3	2	3	2	3	3

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Jurafsky, Daniel, and James H. Martin. Speech and Language Processing. 3rd ed., 2025.
2. Bird, Steven, Ewan Klein, and Edward Loper. Natural Language Processing with Python. 1st ed., 2009.
3. Manning, Christopher D., and Hinrich Schütze. Foundations of Statistical Natural Language Processing. 1999.
4. Goyal, Palash, Sumit Pandey, and Karan Jain. Deep Learning for Natural Language Processing. 1st ed., 2018.
5. Harris, Mary Dee. Introduction to Natural Language Processing. Reston Pub Co, 1985.

Course Contents

Unit 1: Basics Probability The concept of probability, The axioms of probability, theorems on Probability, Basic probability concepts, Conditional Probability, Theorems on conditional probability, Independent Event's, Bayes' Theorem.

Unit 2: Random Variables and Distribution Functions: Random variables, Distribution functions for discrete random variables, Continuous probability distribution, Distributions for continuous random variables, Two Dimensional Random variables, joint distributions, Independent random variables.

Mathematical Expectation: Definition, Functions of random variables, some theorems on Expectation, the variance and Standard Deviation, Moments, Moment Generating Functions, Covariance, Correlation Coefficient.

Unit 3: Probability Distributions: Discrete probability distributions Uniform, Bernoulli, Binomial and Poisson Distribution; Continuous probability distributions Uniform, normal (Gaussian), and exponential distributions, Relations between different distributions

Bayesian Statistics: Bayesian inference: prior, likelihood, and posterior distributions, Bayesian estimation.

Unit 4: Basic of Sampling Distributions: Population and Sample, Sampling with and without replacement, the sample mean, Sampling distribution of means Central limit theorem and student t distribution, Paired-t distribution and distributions of variance Chi-square Distribution and F-distribution, proportions, differences and sums, the sample variance, maximum likelihood estimation (MLE).

Unit 5: Tests of Hypotheses and Significance and advance topics: Statistical Decisions, Statistical hypotheses, Null Hypotheses, Tests of hypotheses and significance, Type I and Type II errors, level of significance, Tests involving the Normal distribution, One-Tailed and Two-tailed tests, Interval estimation, Special tests of significance for large and small samples, The Chi-square test for goodness of fit, P-Values in Hypothesis.

Applications of Probability Theory in Data Analytics Probabilistic models in data analytics, Probability interpretation of regression and correlation, Case studies and applications in various domains

Course Outcomes

CO1: Develop a solid foundation in probability theory and its applications in data analytics.

CO2: Understand the principles and properties of probability distributions and random variables.

CO3: Apply statistical techniques for data analysis, inference, and hypothesis testing.

CO4: Apply probability theory to model uncertainty and make informed decisions in data-driven contexts

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2				2								1	1
CO2	1	2		2	2	2						2	2	3
CO3	1	3		2		1						2		3
CO4	2	2	3	3	2	3						3	2	3

Recommended Books

1. Walpole, R. E., Myers, R. H., Myers, S. L., & Ye, K. (2020). *Probability & Statistics for Engineers & Scientists* (10th ed.). Pearson. ISBN: 9780134115856
2. Devore, J. L. (2015). *Probability and Statistics for Engineering and the Sciences* (9th ed.). Cengage Learning. ISBN: 9781305251809
3. Blitzstein, J. K., & Hwang, J. (2019). *Introduction to Probability* (2nd ed.). CRC Press. ISBN: 9781138369917
4. Wasserman, L. (2004). *All of Statistics: A Concise Course in Statistical Inference*. Springer. ISBN: 9780387402727
5. Myers, R. H., Myers, S. L., & Ye, K. (2012). *Classical and Modern Regression with Applications* (2nd ed.). Duxbury Press. ISBN: 9780534405141

List of Experiments

1. Hands-on practice in creating and modifying database tables using Data Definition Language (DDL) and Data Manipulation Language (DML) commands.
2. Implementation and enforcement of data integrity constraints such as Primary Key, Foreign Key, Unique, and Not Null constraints.
3. Writing and executing SQL queries involving aggregate functions, grouping data, and sorting results using Group By, Having, and Order By clauses.
4. Applying relational algebra operations like UNION, INTERSECT, and MINUS in SQL queries for data manipulation.
5. Constructing SQL queries to retrieve data from multiple tables using various types of joins including Equi-Join, Non-Equi-Join, and Outer Join.
6. Formulating SQL queries containing subqueries and nested queries for complex data retrieval.
7. Utilizing advanced SQL query techniques involving subqueries, set operators (ANY, ALL, IN), and constraints for comprehensive data analysis.
8. Creating and managing database views, indexes, and sequences for efficient data access and manipulation.
9. Practicing Data Control Language (DCL) and Transaction Control Language (TCL) commands for managing database security and transactions.
10. Introduction to PL/SQL programming basics, including variables, control structures, and basic query construction.
11. Writing PL/SQL code with CASE statements, loops, and exception handling mechanisms for effective database programming.
12. Developing stored procedures and user-defined functions in PL/SQL for modular code reuse and customization.
13. Designing and implementing database triggers for automatic actions in response to database events.
14. Understanding and utilizing cursors in PL/SQL for iterative processing of query results.
15. Integrating SQL statements within host programming languages for seamless database interaction.

**** Students are advised to use Oracle/MySQL or other latest database software for above-listed experiments.***

**** Mini Project may also be planned & carried out throughout the semester to understand the important various concepts of Database.***

**** This is only the suggested list of Practical's. Instructor may frame additional Practical's relevant to the course contents.***

Course Outcomes

- CO1: Apply DDL and DML commands proficiently, ensuring data integrity with various constraints.
- CO2: Execute advanced SQL queries with aggregate functions and subqueries.
- CO3: Design and implement database optimizations for performance and security using views, indexes, and sequences.
- CO4: Apply PL/SQL fundamentals for effective transaction management.
- CO5: Utilize Embedded SQL in host languages, showcasing advanced database skills through a mini-project.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PSO 2
CO1	2	3	1	2	2	1					3		3	1
CO2			3	3	2		1				2			
CO3		3	2	2	2			2	2				1	
CO4		2		1	3		2	3	2		1	2		
CO5		1	1	2	2				1	1		2		2

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Implement and classical sorting algorithms (insertion sort, selection sort, bubble sort) and advanced algorithms (merge sort, quick sort, heap sort).
2. Implement linear time sorting algorithms (counting sort, radix sort, bucket sort) and order statistics algorithms (medians and order statistics).
3. Implement divide and conquer algorithms such as binary search, merge sort, quick sort, closest pair problem, and Strassen's matrix multiplication algorithm.
4. Implement and analyze greedy algorithms including fractional Knapsack problem, Huffman coding, Job scheduling, Dijkstra's algorithm, optimal merge pattern, Articulation point.
5. Implement dynamic programming algorithms using memoization and tabulation techniques for problems such as the 0-1 knapsack, longest common subsequence, convex hull and matrix chain multiplication.
6. Implement backtracking algorithms for combinatorial problems such as graph coloring, set generation, permutations, cliques, Hamiltonian cycles, and N Queen.
7. Implement branch and bound algorithms for optimization problems such as the 0/1 knapsack problem, job assignment problem, and traveling salesman problem.
8. Study and analyze complexity theory concepts including P vs NP problem, NP-completeness, NP-hardness, and reduction techniques such as Cook's theorem.
9. Implement various advanced data structures such as Red-Black trees, Binomial Heaps, and Fibonacci Heaps. Perform operations like insertion, deletion, and search, and analyze their time complexities.

****Mini Project may also be planned & carried out throughout the semester to understand the important various concepts of Design and Analysis of Algorithms.***

****This is only the suggested list of Practical's. Instructor may frame additional Practical's relevant to the course contents.***

Course Outcomes

CO1: Develop practical skills in implementing algorithms and data structures.

CO2: Understand the efficiency and correctness of algorithms through empirical analysis.

CO3: Develop experience in analyzing and evaluating algorithm performance using experimental methods.

CO4: Apply algorithmic problem-solving techniques to solve real-world problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	3	3	1	1							3	3	2
CO2	2	3	3	3	2						1	2	2	3
CO3	2	2	2	2	2				1		3	2	2	3
CO4	2	3	3	3	2				1		3	3	3	3

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Hands-on with Network Physical Components (Network Cabling, cable pairing)
2. Introduction to Cisco Packet Tracer/GNS3/Wireshark/NetSim/Mininet/Network Simulator/OPNET – Cover network devices, cables, and end-devices.
3. Connect a PC to another PC and design a network using a hub/switch. Assign IP addresses and subnet masks. Implement network topologies.
4. Topology Implementation: Implement Bus, Ring, Mesh, and Hybrid topologies.
5. Implementation of error detection and correction techniques.
6. Implementation of ARQ protocols such as Stop-and-Wait, Go-Back-N and Selective Repeat.
7. Assign IP addresses and subnet masks, use the ping command for connectivity checks, and observe message flow and configuration status.
8. Network Design Using Routers: Connect two separate networks using a router.
9. Route Configuration: Design static, dynamic, and default route configurations.
10. NAT and PAT Design: Configure Network Address Translation (NAT) and Port Address Translation (PAT).
11. VLAN Configuration: Design and implement VLANs.
12. Real-world Network Infrastructure: Experience the setup and management of real-world network infrastructure.
13. Network Programming – IP Handling, Sending Packets, and UDP/TCP Sockets
14. Client/Server Application Implementation
15. Packet Capturing and Analysis: Use monitoring tools for packet analysis.
16. Application Testing: Use network monitoring tools for application testing.
17. Scripting especially Linux/Unix environment may be encouraged.

** This is only the suggested list of practicals. The instructor may frame additional practicals relevant to the course contents.*

Course Outcomes

- CO1: Understand the processes involved in setting up and configuring network devices, cables, and end-end-devices using simulation tools
- CO2: Apply routing configurations to connect separate networks and design route configurations, including static, dynamic, and default routes.
- CO3: Create comprehensive network designs that incorporate routers, NAT and PAT, and VLAN configurations, and simulating real-world network infrastructures.
- CO4: Apply knowledge of network programming (IP handling, sending packets, UDP/TCP sockets) to develop and test network applications.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3				3								2	3
CO2	3		3	2									3	
CO3	3		3		3									3
CO4	2		3	2	3								3	3

1 – Low, 2 – Medium, 3 – High

List of Experiments

1. Set up and install Python environment for NLP tasks along with NLTK.
2. Implementation of text and character tokenization using NLTK.
3. Write a regex pattern to extract email addresses from a given text.
4. Create a regex to find phone numbers in various formats.
5. Develop a regex to extract dates in different formats (e.g., DD/MM/YYYY, MM-DD,YYYY).
6. Implement character tokenization using `regex_tokenize`.
7. WAP to split a text into words (ignoring punctuation) using regex.
8. Write a regex to remove HTML tags from a string.
9. Implement tokenization for non-English text.
10. Explore tokenization challenges in social media text.
11. Import the Porter's Stemmer and implement stemming to find the stem word.
12. Stem the following words using both Porter and Lovin's stemmers and compare the results: running, flies, easily, organization, troubled, generalization.
13. Tokenize and stem the following sentence using both stemmers: "The cats were running quickly through the structure."
14. Implement and apply various stemming algorithms to a sample text. (i.e. Dawson Stemmer, Krovetz Stemmer, Xerox Stemmer, N-Gram Stemmer Dawson Stemmer, Krovetz Stemmer, Xerox Stemmer, N-Gram Stemmer).
15. Implementing Lemmatization using NLTK, focusing on how it differs from stemming.
16. Implementing Lemmatization using spaCy.
17. Implement Rule-based Part-of-speech (POS) tagging for NLP in Python code using 'averaged_perceptron_tagger'.
18. Implement NER using SpaCY in python using NLTK with Pre-Trained Models.
19. Write a Python program that takes a user-input sentence and parses it using constituency parsing with NLTK. The program should generate a parse tree using a given CFG. Visualize the generated parse tree using NLTK's tree visualization tools.

**This is only the suggested list of Practicals. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

- CO1: Understand fundamental concepts in natural language processing, including syntax, semantics, pragmatics, and discourse analysis.
- CO2: Apply preprocessing techniques such as tokenization, stemming, lemmatization and part-of-speech tagging to raw data.
- CO3: Design and implement information extraction system to extract structured information from unstructured text data.
- CO4: Build basic machine translation system using statistical and neural machine translation techniques.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3		3										3	2
CO2	3		3	3	2						1		2	
CO3	3	2		2					1		2		3	3
CO4	2	3	3		2				1			1		2

1 – Low, 2 – Medium, 3 - High

5th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0301	Artificial Intelligence	3	0	0	3	DC
2	CSDC0305	Operating Systems	3	0	0	3	DC
3	CSDC0307	Data Mining and Analytics	3	0	0	3	DC
4	CSDC0309	Computer Vision	3	0	0	3	DC
5	CSDE03XX/ XXGE03XX	Department Elective – I/ Generic Elective - I	3	0	0	3	DE/GE
6	CSDC0335	Operating Systems Laboratory	0	0	2	1	DC
7	CSDC0337	Data Mining and Analytics Laboratory	0	0	2	1	DC
8	CSDC0339	Computer Vision Laboratory	0	0	2	1	DC
9	CSTR0300	Industrial Practical Training (after 4thSem)	0	0	0	2	TR
10	CSPR0301	Project-1 (Phase-I)	0	0	2	2	PR
TOTAL			15	0	8	22	

Course Contents

Unit 1: Introduction to Artificial Intelligence

Definitions, The foundations of Artificial Intelligence, The History of Artificial Intelligence, The state of the art. Intelligent: Agents and Environments, The concept of Rationality, The Nature of Environments, The Structure of Agents.

Unit 2: Problem Solving

Solving Problems by Searching: Problem-Solving Agents, Example Problems, Searching for Solutions, Uninformed Search, Informed (Heuristic Search), Heuristic Functions. Hill climbing search, Adversarial search: Game Playing Optimal decision in Games, Mini-max search procedure, Alpha-beta cut-offs, Constraint Satisfaction Problems.

Unit 3: Knowledge Representation

Logical Agents, Propositional Logic, First order Logic, Inference in First Order Logic: Unification and Resolution.

Unit 4: Uncertain Knowledge and Reasoning

Quantifying Uncertainty- Acting under uncertainty, Inference using full joint Distributions, Independence, Bayes Rule. Probabilistic reasoning, representing knowledge in uncertain domain, the semantics of Bayesian networks, Exact Inference in Bayesian Networks, Approximate Inference in Bayesian Networks

Unit 5: Case Studies & Trends in AI

Landmark AI case studies (AlphaFold, ChatGPT, AlphaGo, autonomous vehicles, AI in healthcare), AI in industries (finance, marketing, cybersecurity, legal tech), ethical & societal issues (bias, XAI, misinformation, privacy), and emerging trends (multimodal AI, neuroscience, sustainability, generalist AI models).

Course Outcomes

CO1: Analyse and evaluate problems to determine their suitability for AI based solutions

CO2: Apply AI algorithms to formalize a given problem and represent knowledge using various languages/frameworks

CO3: Understand the fundamental principles underlying various AI algorithms, such as standard search algorithms or resolution

CO4: Analyse and apply modern tools and technologies to solve real-world problems

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	2		3		3	3		2		2		2	1
CO2	2	3	2	2	3	1							2	2
CO3	2		3		2								1	1
CO4				2	3				3			3	2	1

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Russell, Stuart, and Peter Norvig. Artificial Intelligence: A Modern Approach. 4th ed., Prentice Hall, 2020.
2. Michalski, Ryszard S., Jaime G. Carbonell, and Tom M. Mitchell. Machine Learning: An Artificial Intelligence Approach. Morgan Kaufmann, 1983.
3. Alpaydin, Ethem. Introduction to Machine Learning. 2nd ed., MIT Press, 2010.
4. Mitchell, Tom. Machine Learning. 1st ed., McGraw Hill Education, 2017.
5. Biere, Armin, Marijn Heule, Hans van Maaren, and Toby Walsh, editors. Handbook of Satisfiability. IOS Press, 2009.

Course Contents**Unit 1: Operating Systems Objectives, Services, and Functions**

Introduction to operating system, Types of operating systems and differences among them; Basic Computer Architecture, OS as a virtual machine; User and Operating System View, System Calls, System Services, Linkers and Loaders, Booting, OS as a resource manager, Interrupts and traps, System calls, Limited direct execution, user versus kernel mode.

Unit 2: Processes and Deadlocks

Introduction to processes management, various process transition states, Introduction to Processor scheduling, types of schedulers, Performance criteria in scheduling algorithms, FCFS scheduling algorithm, Priority scheduling algorithm like SJF, Non-preemptive and Preemptive algorithms, Round-robin scheduling algorithm, Concept of multi-level queues, feedback queues,, Real- Time CPU Scheduling, Thread v/s process, Process and Thread APIs, Process Synchronization: Inter-process communication and Processes: IPC in Shared-Memory Systems and Message-Passing Systems, Race condition, mutual exclusion, The Critical Section Problem (CSP), Algorithmic solutions to CSP – Dekker's, Peterson's, Lamport Bakery Solution; Hardware Support for Synchronization – Test and Set, Compare and Swap; OS support for synchronization - Mutex Locks, Semaphores, Monitors; Condition Variables; Classic Problems of Synchronization – Producer Consumer, Sleeping Barber; Dining Philosopher's Problem, Deadlocks: Deadlocks, issues related to its prevention, avoidance and detection / recovery,

Unit 3: Memory Management

Need of Memory management and its requirements, paging, segmentation, fragmentation. Characteristics of contiguous & non-contiguous allocation techniques, Virtual memory management,, Paging Techniques, FIFO and optimal page-replacement algorithms, Anomalies, Segmentation Techniques and combination with Paging

Unit 4: I/O Management

Overview of Mass-Storage Structure, HDD Scheduling,,; I/O Systems -Overview; I/O Hardware; Kernel I/O Subsystem, Transforming I/O Requests to Hardware Operations.

Unit 5: File Management

File Concept, Access Methods, Directory Structure, Protection, File System Interface, Shared files. File-System Implementation: Structure and Operations; Directory Implementation; Allocation Methods; Free-Space Management; Case study: EXT, NTFS, HFS.

Course Outcomes

- CO1: Understand the fundamental functions and components of an operating system.
- CO2: Compare and contrast different operating system designs and algorithms, and assess their suitability for various scenarios.
- CO3: Examine the operational mechanisms of various subcomponents within the operating system.
- CO4: Demonstrate proficiency in implementing programming and system-level tasks within an operating system.
- CO5: Apply problem-solving strategies and critical-thinking skills in the context of operating systems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	1	3												
CO2		3	2	1									1	1
CO3		2	2	3	1									
CO4		2	3	3	3								3	
CO5		3	3	3	2								2	1

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Peterson, Abraham, and Galvin Silberschatz. Operating System Concepts. 10th ed., Addison-Wesley, 2023.
2. Tanenbaum, Andrew S. Modern Operating Systems. 5th ed., Pearson Education, 2024.
3. Stallings, William. Operating Systems. 9th ed., Pearson, 2018.
4. Dhamdhere, D. M. Introduction to Systems Software. 3rd ed., Tata McGraw-Hill, 2017.
5. Chauhan, N. Principles of Operating Systems. 1st ed., 2018.

Course Contents

Unit 1: Foundations of Data Mining and Analytics

What is Data Mining, Data Mining: An essential step in knowledge discovery, Diversity of data types for data mining, Mining various kinds of knowledge, Data Mining and applications. Data, measurements and data preprocessing, Statistics of data, Similarity and distance measures, Data quality, data cleaning and data integration, Data transformation, Dimensionality reduction. Overview of Data analytics and its importance, Introduction to Data Analytics Life Cycle

Unit 2: Pattern Mining

Basic concepts, Frequent itemset mining methods: Apriori Algorithms, Generating association rules from frequent itemsets, Improving the efficiency of Apriori, Frequent-Pattern growth. Pattern evaluation methods.

Unit 3: Data Warehousing

Data warehouse modelling: schema for multidimensional data models, concept hierarchies, measures: categorization and computations. OLAP operations, Data cube computations

Unit 4: Tools for Data Mining & Analytics

Introduction to MongoDB, Data Types, and MongoDB Query Language. Introduction to Hadoop, RDBMS vs Hadoop, key aspects of Hadoop, Hadoop components

Unit 5: Ethics & Trends in Data Mining & Analytics

Case Study and real-world applications of data mining and analytics, Mining Rich data types, Data Mining Methodologies and systems. Ethical consideration and data privacy, Bias and fairness in data analytics, Ethical challenges and guidelines, Responsible data analytics practices, Emerging Trends in data analytics

Course Outcomes

CO1: To explain core concepts of data mining and analytics, including data types, preprocessing, statistics, and lifecycle.

CO2: To apply mining algorithms like Apriori and FP-Growth for pattern discovery and association analysis.

CO3: To design data warehouse models and use tools like MongoDB and Hadoop for data analytics.

CO4: To evaluate ethical concerns, privacy issues, and emerging trends using case studies.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	2	2	2	2								2	
CO2	3	3	2	2	2									3
CO3	3	3	2	3	3									3
CO4						3		3					2	

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Han, Jiawei, Jian Pei, and Hanghang Tong. Data Mining: Concepts and Techniques. Morgan Kaufmann, 2022.
2. Leskovec, Jure, Anand Rajaraman, and Jeffrey David Ullman. Mining of Massive Data Sets. Cambridge University Press, 2020.
3. Acharya, Seema, and Subhashini Chellappan. Big Data and Analytics. Wiley, 2019.
4. Black, Ken. Business Statistics: For Contemporary Decision Making. John Wiley & Sons, 2023.
5. Russell, Matthew A. Mining the Social Web: Data Mining Facebook, Twitter, LinkedIn, Google+, GitHub, and More. O'Reilly Media, 2013.

Course Contents**Unit 1: Introduction**

Digital Image Formation and low-level processing, Steps of Image Processing, Fundamentals of Image Formation, The Four R's of Computer Vision, Geometry of Image Formation, ssampling and quantization Neighborhood Processing, Connectivity and relation between the pixels.

Unit 2: Edge and Object Detection

Image Enhancement, Restoration, Histogram Processing: Histogram equalization, Histogram matching, Filtering: Smoothing filters (mean, Gaussian), Sharpening filters (Laplacian, gradient-based), Concepts in Edge Detection: Sobel, Prewitt, Roberts, Canny, Line detectors (Hough Transform), Corners - Harris and Hessian Affine, Concepts of Object Detection: Haar Cascades, Histogram of Oriented Gradients (HOG).

Unit 3: Segmentation & Feature Extraction

Segmentation: active contour, split and merge, region growing Feature extraction: GLCM, LBP, SIFT, SURF and BRISK features, Scale-Space Analysis-Image Pyramids and Gaussian derivative filters, Texture analysis using Gabor Filters, Concept of object recognition, classification and detection, Representation schemes like chain coding, Signatures, Boundary Descriptors like Shape Numbers.

Unit 4: Machine Learning for Computer Vision

Overview of learning-based methods applied to visual data interpretation, Dataset preparation: characteristics of annotated visual samples, Techniques for data sampling, augmentation, and enhancement to improve model training, Visual content categorization using image classifiers, Pixel-wise labelling and region classification through semantic segmentation methods, Evaluation of model effectiveness using classification metrics.

Course Outcomes

- CO1: Understand the fundamental concepts of computer vision and image processing.
- CO2: Apply and analyse the various image enhancement, restoration and edge detection techniques.
- CO3: To implement segmentation and feature extraction methods such as GLCM, LBP, SIFT, SURF, and shape descriptors for object recognition and classification.
- CO4: Evaluate different machine learning techniques, including supervised and unsupervised methods, for tasks like image classification, object detection, and semantic segmentation using performance metrics.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	2	1									1	1	
CO2	2	2	2									2	1	
CO3	3	3	2	2							1	2	1	
CO4	3	3	3	3							1	2	2	

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Rafael C. Gonzales and Richard E. Woods, Digital Image Processing, Pearson Education, 3rd Edition, 2007.
2. Milan Sonka, Vaclav Hlavac, and Roger Boyle, Digital Image Processing and Computer Vision, Cengage Learning, 2007.
3. Szeliski, R., Computer Vision: Algorithms and Applications, Springer Nature, 2022.
4. D. Forsyth and J. Ponce, Computer Vision: A Modern Approach, 2nd Edition, Pearson.
5. B. Chanda, Digital Image Processing and Analysis, PHI Learning Pvt. Ltd., 2011.
6. William K. Pratt, Digital Image Processing, Wiley-Interscience, 4th Edition, 2007.

List of Experiments

1. Installation and working with GCC on Linux
2. Write a shell script that performs the following tasks:
 - a) File manipulation: Create, delete, copy, and move files and directories.
 - b) Text processing: Search for specific patterns in files and perform text transformations.
 - c) System monitoring: Retrieve system information like CPU usage, memory utilization, and disk space.
 - d) Automation: Automate a repetitive task on your Linux system using a shell script.
3. Develop a program that interacts with the Linux file system. Your program should enable users to:
 - a) Create files and directories.
 - b) Navigate through directories and display their contents.
 - c) Copy or move files and directories.
 - d) Change file permissions and ownership.
4. Implement a program in Linux that demonstrates the following process management concepts:
 - a) Process creation: Create child processes using the fork() system call.
 - b) Process termination: Terminate processes using the exit() system call.
 - c) Process synchronization: Synchronize processes using semaphores, mutexes, or other synchronization mechanisms.
 - d) Signal handling: Handle signals like SIGINT or SIGTERM in your program
5. Simulation of the CPU scheduling algorithms such as Round Robin, SJF, FCFS, Priority
6. Simulation of MUTEX and SEMAPHORES.
7. Simulation of Bankers Deadlock Avoidance and Prevention algorithms.
8. Implementation of Process Synchronization (Reader-Writer, Sleeping Barber and Dining Philosopher's Problem)
9. Simulation of page Replacement Algorithms such as FIFO, LRU, and Optimal
10. Simulation of paging techniques of memory management.
11. Simulation of Disk Scheduling Algorithms such as SSTF, SCAN, LOOK etc.
12. Simulation of file allocation Strategies like Sequential, Indexed, Linked
13. Simulation of file organization techniques such as Single Level Directory, Two Level, Hierarchical, DAG.

** This is only the suggested list of Practicals. Instructor may frame additional relevant Practical*

Course Outcomes

- CO1: Analyse the working of an operating system and its components and define and analyse the synchronization process.
- CO2: Identify the working methodology of multithreaded applications.
- CO3: Compare and analyse different file systems being used in different operating systems.
- CO4: Design and implement main memory and secondary memory systems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	3		2				2	2					
CO2		2	3	1									2	
CO3		2		3	2		2						1	
CO4	1	1		2	2								1	1

1 – low, 2 – Medium, 3 - High

List of Experiments

1. Installation of R and Perform the basic operations in R.
2. Perform various operations on data structures in R.
3. To perform different operations on data frames.
4. To explore various built-in R functions for descriptive statistics (mean, median, variance, standard deviation, percentile, quartile, skewness, kurtosis, etc) and probability distributions (binomial, normal, uniform, chi-square, Poisson, etc.).
5. Load a built-in dataset in R, manipulate and analyse the data using dplyr functions and apply techniques for clean and pre-processing data by normalising, transforming, and handling missing values.
6. Conducting feature selection and filtering on large datasets using PCA or decision trees.
7. To know and implement different types of regressions. Perform Logistic Regression on any suitable dataset.
8. To perform classification on any suitable dataset using a decision tree, naive Bayes, known and SVM algorithms.
9. Implementing the Apriori algorithm to identify frequent item sets in a dataset.
10. Implementing and comparing various clustering techniques (K-means, hierarchical, DBSCAN) on given datasets.
11. Install Hadoop, perform the given file management, and configure tasks.
12. Using MongoDB, create a collection hotel with the given document & demonstrate the CRUD operations.

** This is only the suggested list of Practicals. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

- CO1: Understand and implement data pre-processing and exploratory data analysis techniques, enabling effective visualization and identification of patterns in data for insightful analytics.
- CO2: Understand descriptive statistics, including measures of central tendency and dispersion, and master basic and advanced statistical analysis techniques.
- CO3: Apply machine learning algorithms for predictive modeling and data mining.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3			3	3								2	3
CO2	2				3								1	3
CO3	3			2	3								3	3

1 – low, 2 – Medium, 3 - High

List of Experiments

1. Write a program to read an input image in Python.
2. Write a program to compute Mean, Median, Variance and Standard Deviation of the given Image.
3. Write a program to implement contrast stretching of a low contrast image, Histogram, and Histogram
4. Equalization.
5. Installation of Matlab and Computer Vision toolbox. Write a program to implement any algorithm for image segmentation.
6. Write a program to understand and implement Canny Edge detection algorithm.
7. Write a program to implement edge detection using Sobel operator.
8. Write a program to implement SIFT, SURF, BRISK feature descriptor an image.
9. Write a program to implement histogram thresholding segmentation.
10. Write a program, to use morphological filters on binary image only.
11. Write a program to perform classification on images using a suitable machine learning model.
12. Project: Implement a Convolutional Neural Network architecture and perform 4-class classification on the CIFAR-10 dataset. Also evaluate the trained CNN model using metrics such as Precision, Recall and Accuracy.

**This is only the suggested list of Practicals. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

CO1: Understand the different computer vision algorithms.

CO2: Understand and gain experience on different computer vision tools.

CO3: Apply programming and scientific tools for relevant software implementation.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO1 2	PSO 1	PSO 2
CO1		3					2						2	2
CO2	1			2										2
CO3		2				3								2

1 – Low, 2 – Medium, 3 - High

List of Experiments

1. Write a program to read an input image in Python.
2. Write a program to compute Mean, Median, Variance and Standard Deviation of the given Image.
3. Write a program to implement contrast stretching of a low contrast image, Histogram, and Histogram
4. Equalization.
5. Installation of Matlab and Computer Vision toolbox. Write a program to implement any algorithm for image segmentation.
6. Write a program to understand and implement Canny Edge detection algorithm.
7. Write a program to implement edge detection using Sobel operator.
8. Write a program to implement SIFT, SURF, BRISK feature descriptor an image.
9. Write a program to implement histogram thresholding segmentation.
10. Write a program, to use morphological filters on binary image only.
11. Write a program to perform classification on images using a suitable machine learning model.
12. Project: Implement a Convolutional Neural Network architecture and perform 4-class classification on the CIFAR-10 dataset. Also evaluate the trained CNN model using metrics such as Precision, Recall and Accuracy.

**This is only the suggested list of Practicals. The instructor may frame additional Practical relevant to the course contents.*

Course Outcomes

CO1: Understand the different computer vision algorithms.

CO2: Understand and gain experience on different computer vision tools.

CO3: Apply programming and scientific tools for relevant software implementation.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO1 2	PSO 1	PSO 2
CO1		3					2						2	2
CO2	1			2										2
CO3		2				3								2

1 – Low, 2 – Medium, 3 - High

6th Semester

S. No	Course Code	Course Title	Teaching Load			Credit	Course Type
			L	T	P		
1	CSDC0304	Information Security Systems	3	0	0	3	DC
2	CSDC0306	Machine Learning	3	0	0	3	DC
3	CSDC0308	Software Engineering	3	0	0	3	DC
4	CSDC0310	Deep Learning	3	0	0	3	DC
5	CSDE03XX/ XXGE03XX	Department Elective – 2/ Generic Elective - 2	3	0	0	3	DE/GE
6	CSDC0334	Information Security System Laboratory	0	0	2	1	DC
7	CSDC0336	Machine Learning Laboratory	0	0	2	1	DC
8	CSDC0338	Software Engineering Laboratory	0	0	2	1	DC
9	CSDC0340	Deep Learning Laboratory	0	0	2	1	DC
10	CSPR0302	Project-1 (Phase-II)	0	0	2	2	PR
TOTAL			15	0	10	21	

Course Contents**Unit 1: Introduction to Computer Security and Information Security**

Computer Security: Introduction to Computer Security, Computer Security Technology and Principles, Software Security and Trusted Systems, Networking and Distributed System Security, Information Security: History and Evolution of Cryptography, OSI Security Architecture, A Model for Network Security, Classical Encryption Techniques: Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Steganography, Block Ciphers and Data Encryption Standard: Block Cipher Principles, Simplified DES, Data Encryption Standard, Strength of DES, Block Cipher Modes of Operation, confusion and diffusion & their measurement.

Unit 2: Mathematical Foundation and Symmetric Key Cryptography

Introduction to Finite Fields: Groups, Rings, and Fields, Modular Arithmetic, Euclid's Algorithm, and Forms of Finite Fields, Advanced Encryption Standard: Evaluation Criteria for AES, AES Cipher, Contemporary Symmetric Ciphers: Triple DES, Blowfish, Characteristics of Advanced Symmetric Block Ciphers, RC4 Stream Cipher, Confidentiality Using Symmetric Encryption: Placement of Encryption Function, Traffic Confidentiality, Key Distribution, Random Number Generation.

Unit 3: Asymmetric Key Cryptography

Public-Key Encryption: Introduction to Number Theory: Prime Numbers, Fermat's and Euler's Theorems, Testing for Primality, Chinese Remainder Theorem, Public-Key Cryptography and RSA: Principles of Public-Key Cryptosystems, RSA Algorithm, Key Management and Other Public-Key Cryptosystems: Key Management, Diffie-Hellman Key Exchange, Elliptic Curve Arithmetic, Elliptic Curve Cryptography.

Unit 4: Message Authentication and Hash Functions

Message Authentication and Hash Functions: Authentication Requirements, Authentication Functions, Message Authentication Codes, Hash Functions, Security of Hash Functions and MACs, Hash Algorithms: Message Digest Algorithm (MD5), Secure Hash Algorithm, RIPEMD-160, and HMAC, Digital Signatures and Authentication Protocols: Digital Signatures, Authentication Protocols, Digital Signature Standard.

Unit 5: Cryptanalysis

Cryptanalysis: Time-Memory Trade-off Attack, Cryptanalysis of the Substitution Cipher, Cryptanalysis of the Hill Cipher, Cryptanalysis of the LFSR Stream Cipher, Cryptanalysis on Stream Cipher, Modern Stream Ciphers, Shamir's Secret Sharing, Differential and Linear Cryptanalysis.

Course Outcomes

- CO1: Understand fundamental principles of cryptography, including Integrity, Confidentiality, authentication, and Non-repudiation, to secure communication systems.
- CO2: Analyse security issues and vulnerabilities in networks and cryptographic systems
- CO3: Assess network security threats and develop effective countermeasures, implementing secure network architectures using cryptographic techniques.
- CO4: Evaluate security mechanisms, including key ciphers and hash functions, using rigorous mathematical and computational approaches for cryptographic security.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	3	1	1		1					2	1	2	
CO2	2	2		1	1							1	1	2
CO3		2	1	3	2	2					2	2	2	2
CO4	1	2	2		2							1	2	

1 – Low, 2 – Medium, 3 - High

Recommended Books

1. Stallings, William. Cryptography and Network Security: Principles and Practice. 8th ed., Pearson Education, 2024.
2. Forouzan, Behrouz A., and Debdeep Mukhopadhyay. Cryptography and Network Security. 3rd ed., Tata McGraw-Hill, 2015.
3. Boneh, Dan, and Victor Shoup. A Graduate Course in Applied Cryptography. Version 0.6, Jan. 2023.
4. Stinson, Douglas R., and Maura Paterson. Cryptography: Theory and Practice. 4th ed., Chapman & Hall/CRC, 2018.
5. Menezes, Alfred J., Paul Van Oorschot, and Scott Vanstone. Handbook of Applied Cryptography. CRC Press, 2018.
6. Gupta, Prakash C. Cryptography and Network Security. 1st ed., PHI Learning, 2015.

Course Contents**Unit 1: Introduction to Machine Learning and Mathematical Foundations**

Introduction to ML: Definitions, Applications, Types (Supervised, Unsupervised, Reinforcement Learning). History and Evolution of ML, Comparison with Traditional Programming ML Pipeline: Data Collection, Processing, Model Selection, Training, Evaluation.

Linear Algebra: Vectors, Matrices, Eigenvalues & Eigenvectors, Optimization Techniques: Gradient Descent, Convex Optimization, Loss Functions: MSE, Cross-Entropy, Hinge Loss, Evaluation Metrics: Accuracy, Precision, Recall, F1 Score, ROC Curve

Unit 2: Supervised Learning

Regression Models, Linear Regression: Model, Cost Function, Gradient Descent, Multiple Linear Regression & Polynomial Regression Ridge & Lasso Regression

Classification Models Logistic Regression, Decision Trees: Splitting Criteria (Gini Index, Entropy), Pruning of decision tree, Multivariate Trees, Hypothesis space search in decision tree learning, Inductive bias in decision tree learning, Issues in decision tree learning. Random Forests & Ensemble Learning (Bagging, Boosting, Stacking), Support Vector Machines (SVM): Kernel Trick, Hyperplane Concept, k-Nearest Neighbors (k-NN) Algorithm, Naïve Bayes Classifier, Bayesian belief networks Case Study: Predicting House Prices and Spam Classification, Artificial Neural Network (ANN).

Unit 3: Unsupervised Learning

Introduction to Unsupervised Learning, Clustering: K-Means, k-medoid Hierarchical Clustering, DBSCAN, Clustering & Gaussian Mixture Models (GMM), Case Study: Customer Segmentation & Anomaly Detection.

Unit 4: Reinforcement Learning

Overview of Reinforcement Learning: Elements of Reinforcement Learning, Markov Decision Process (MDP) and its Components, Exploration vs. Exploitation Trade-off, Dynamic Programming for RL: Bellman Equations, Value Iteration and Policy Iteration, Finding Optimal Policies, Model-Free Reinforcement Learning, Temporal Difference (TD) Learning: SARSA (On-Policy Learning) Q-Learning (Off-Policy Learning), REINFORCE Algorithm and Actor-Critic Methods (A2C, A3C), Exploration Strategies: ϵ -Greedy, Softmax Exploration, Applications of Reinforcement Learning.

Course Outcomes

CO1: Understand fundamental machine learning concepts and their real-world applications.

CO2: Implement and analyze machine learning algorithms, including regression, classification, clustering, and reinforcement learning.

CO3: Evaluate machine learning models using metrics like accuracy, precision, recall, and F1-score.

CO4: Apply machine learning techniques to real-world problems, optimize models, and interpret results for better decision-making.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2		1						1		1	2	2
CO2	3	3	3	3	2								3	1
CO3	2	2	1	3						1	3	2	2	2
CO4	2	2	2	2					1		3	2		1

1 – Low, 2 – Medium, 3 - High

Recommended Books

1. Mohri, Mehryar, Afshin Rostamizadeh, and Ameet Talwalkar. Foundations of Machine Learning. 2nd ed., MIT Press, 2018.
2. Sammut, Claude, and Geoffrey I. Webb, editors. Encyclopedia of Machine Learning and Data Mining. 3rd ed., Springer, 2023.
3. Witten, Ian H., Eibe Frank, Mark A. Hall, and Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques. 5th ed., Morgan Kaufmann, 2024.
4. Müller, Andreas C., and Sarah Guido. Introduction to Machine Learning with Python: A Guide for Data Scientists. O'Reilly Media, 2016.
5. Bishop, Christopher M. Pattern Recognition and Machine Learning. Springer, 2013.

Course Contents**Unit 1: Introduction to Software engineering**

Software engineering approach, the evolving role of software, Software myths, Software process, Characteristics of software process, Software development process models.

Unit 2: Software requirements specification

Software requirements, problem analysis, requirements specification, functional specification with use cases, validation.

Unit 3: Design Concepts

Abstraction, Refinement, Modularity, and Architectural design. Object-Oriented Approach: Concepts, Design methodology, Design issues, UML, Design patterns. User interface design: User interface analysis and design, Interface design steps, Design evaluation.

Unit 4: Quality Management

Software quality assurance, Software configuration management, Software testing strategies, Black-Box and White-Box testing, Validation testing, System testing, designing test cases, Debugging, Bug tracking, CASE tools, Product metrics

Unit 5: Introduction to Agile Development

Agile Methodologies, Agile Practices, Agile Project Management, Agile Estimation and Budgeting, Agile Scheduling and Risk Management, Agile Tools and Case Studies

Unit 6: Advanced Topics

Software process improvement, emerging trends in software engineering

Course Outcomes

CO1: Describe the concepts of software development and software engineering.

CO2: Differentiate between various software engineering process models and explain their applicability.

CO3: Implement testing and debugging techniques and evaluate their effectiveness.

CO4: Assess software risks, apply risk management strategies, and analyze software quality and reliability based on international quality standards.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	1	2	2				1						1	
CO2		2		3				2					2	
CO3			3	2	3								2	2
CO4		3			3	3	2		3			3	3	3

1 – Low, 2 – Medium, 3 - High

Recommended Books

1. Peters, James F., and Witold Pedrycz. Software Engineering: An Engineering Approach. 1st ed., John Wiley & Sons, 2000.
2. Jawadekar, Waman S. Software Engineering: Principles and Practice. 1st ed., McGraw-Hill Education, 2004.
3. Pressman, Roger S., and Bruce R. Maxim. Software Engineering: A Practitioner's Approach. 9th ed., McGraw-Hill Education, 2019.
4. Sommerville, Ian. Software Engineering. 10th ed., Pearson Education, 2015.
5. Jalote, Pankaj. Software Engineering: A Precise Approach. John Wiley & Sons, 2010.
6. Fox, Armando, David A. Patterson, and Samuel Joseph. Engineering Software as a Service: An Agile Approach Using Cloud Computing. Strawberry Canyon LLC, 2013.

Course Contents

Unit 1: Introduction

Introduction to Deep Learning and its Applications. Introduction to Statistical Learning: Multi-Layer Perceptron, Back Propagation, Linear Regression, Loss Functions and Optimization: Optimization, stochastic gradient descent, dropout, batch normalization, etc.

Unit 2: Convolutional Neural Networks Architecture

Convolution, pooling, Activation Functions, Back propagation of CNN, Weights as templates, Translation invariance, Training with shared parameters, AlexNet, VGG, GoogLeNet, ResNet, Capsule Net, etc., Visualizing intermediate features and outputs, Saliency maps, Visualizing neurons, Cam-Grad, etc.

Unit 3: Sequential Modelling

Recurrent and Recursive Nets, RNN, LSTM, GRU, Image captioning, visual question answering, etc.

Unit 4: Generative Models

Encoder, Decoders, Variational Autoencoders, Generative Adversarial Networks like pix2pix, CycleGAN, etc. Transformers based Models

Unit 5: Applications and Framework

Object Detection: RCNN, Fast RCNN, Faster RCNN, YOLO and variants, Retina Net, etc., Adversarial Attacks on CNN, Deep learning for NLP, Keras, TensorFlow, PyTorch, AutoML, etc

Course Outcomes

CO1: Understand the foundational concepts of deep learning

CO2: Implement and train deep neural networks using popular frameworks.

CO3: Analyze and evaluate the performance of deep learning models.

CO4: Apply deep learning techniques to solve real-world problems.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	3	2	2										3	1
CO2	2	3	3	2	2	1							1	3
CO3	2	3	3	3	2								2	3
CO4		2		2	3	1		2	3				3	3

1 – Low, 2 – Medium, 3 – High

Recommended Books

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.
2. Michael A. Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.
3. Josh Patterson, Deep Learning: A Practitioner's Approach, 1st Edition, O'Reilly Media, 2017.
4. François Chollet, Deep Learning with Python, Manning Publications, 2024.
5. Nithin Buduma, Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, 2nd Edition, O'Reilly Media, 2022.

List of Experiments

1. Implementation of various transposition ciphers.
2. Implementation of various substitution ciphers.
3. Implementation of S-DES algorithm for data encryption.
4. Implementation of S-AES algorithm for data encryption.
5. Implementation of Symmetric Encryption Scheme – Mode of operations and RC4.
6. Implementation of Chinese remainder theorem.
7. Implementation of RSA cryptosystem.
8. Implementation of Rabin cryptosystem.
9. Implementation of ElGamal cryptosystem.
10. Implementation of various Secure Hash algorithms.
11. Implementation of MAC algorithm.
12. Implementation of RSA digital signatures.
13. Implementation of Rabin digital signatures.
14. Implementation of ElGamal digital signatures.
15. Implementation of Diffie-Hellman key exchange algorithms

** This is only the suggested list of Practical's. Instructor may frame additional Practicals relevant to the course contents.*

Course Outcomes

CO1: Understanding of various information security threats.

CO2: Understanding of security measures to encounter threats.

CO3: Implementation and understanding of encryption techniques.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	1	3			2	2	1		1				1	2
CO2		2		1	2	3	2						2	2
CO3	1	2			3				2	2	1	1	2	3

1 – low, 2 – Medium, 3 - High

List of Experiments

1. Programs based on Numpy, Pandas, Visualization based python libraries.
2. Implementation of regression and logistic regression for classification purpose.
3. Implementation of Decision Tree for solving regression as well as classification problems,
4. Solving classification and regression problems with application of Support Vector Machines (SVM) in Data Analytics.
5. Implement Dimensionality reduction using Principle Component Analysis (PCA) method,
6. Implement Naïve Bayes Classification in Python, Build KNN Classification model for a given dataset.
7. Implement various ensemble methods on a given dataset,
8. Implement Random Forest ensemble method on a given dataset,
9. Implement Boosting ensemble method on a given dataset.
10. Implementation of clustering techniques, such as, K-Means clustering Algorithm.
11. Estimate the accuracy of decision classifier on breast cancer dataset using 5-fold cross-validation
12. Develop a machine learning method to Cluster gene expression data, how to modify existing methods to solve the problem better.
13. Select two datasets. Each dataset should contain examples from multiple classes. For training purposes assume that the class label of each example is unknown (if it is known, ignore it). Implement the K-means algorithm and apply it to the data you selected. Evaluate performance by measuring the sum of Euclidean distance of each example from its class center. Test the performance of the algorithm as a function of the parameter k.
14. Implement the EM algorithm assuming a Gaussian mixture. Apply the algorithm to your datasets and report the parameters you obtain. Evaluate performance by measuring the sum of Mahalanobis distance of each example from its class center. Test performance as a function of the number of clusters.

** This is only the suggested list of Practicals. Instructor may frame additional Practical's relevant to the course contents.*

Course Outcomes

CO1: Develop hands-on knowledge of methods and theories in the field of machine learning for learning principles, techniques, and applications of machine learning.

CO2: Design and Implement Regression for real-world problems.

CO3: Design and implement various classification techniques.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1			3	3									2	
CO2					3		2						3	3
CO3			3	2	2				2				1	

1 – Low, 2 – Medium, 3 - High

List of Experiments

1. Introduction and project definition
2. Introduction to project management software (e.g., Trello, Jira)
3. Software process overview (Waterfall, Agile, Scrum, DevOps)
4. Project planning (WBS, Gantt charts, risk management)
5. Software Requirements Specification (SRS)
6. Introduction to UML and use case diagrams
7. Flow of events and activity diagrams
8. Object-oriented analysis: discovering classes and class diagrams
9. Interaction diagrams: sequence and collaboration diagrams
10. Software design: architecture and object-oriented design
11. Effort and cost estimation using COCOMO
12. State transition diagrams
13. Component and deployment diagrams
14. Issue tracking systems (e.g., GitHub, Bitbucket, GitLab)
15. Software configuration management tools (e.g., Git, SVN, CVS)
16. Software testing (Unit, integration, system, acceptance testing)
17. Continuous integration and delivery tools (e.g., Jenkins, GitHub Actions)
18. Writing test cases and using test management tools (e.g., TestLink)
19. Code review practices and tools (e.g., GitHub pull requests)

* This is only the suggested list of Practicals. Instructor may frame additional Practical's relevant to the course contents.

Course Outcomes

CO1: Apply project planning techniques and tools in software project management.

CO2: Use SDLC models and project management software effectively.

CO3: Create UML diagrams and SRS for software requirements.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	P O7	PO8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1		3	2	2	2	2			3	3			1	
CO2					2	3							2	
CO3					3			3	2				2	3

1 – Low, 2 – Medium, 3 - High

List of Experiments

1. Implement a basic MLP using TensorFlow or PyTorch and train the MLP using Back Propagation and various optimization techniques like stochastic gradient descent.
2. Implement a CNN architecture for image classification using TensorFlow or PyTorch and explore different convolutional layers, pooling layers, and activation functions.
3. Study and implement popular CNN architectures such as AlexNet, VGG, GoogLeNet, ResNet, and Capsule Net.
4. Analyze the architectures' design principles and discuss their strengths and weaknesses.
5. Generate saliency maps to understand which parts of the input image contribute most to the predictions.
6. Implement RNN, LSTM, and GRU models for sequential data processing tasks like language modeling or time series prediction.
7. Explore applications like image captioning and visual question answering using RNN-based models.
8. Implement variational autoencoders (VAEs) and Generative Adversarial Networks (GANs) using TensorFlow or PyTorch.
9. Explore applications like image-to-image translation using pix2pix and CycleGAN.
10. Implement transformer-based models like BERT or GPT using pre-trained models available in libraries like Hugging Face Transformers.
11. Fine-tune the pre-trained models on specific tasks like text classification or language generation.
12. Implement object detection algorithms like RCNN, Fast RCNN, Faster RCNN, YOLO, and RetinaNet using TensorFlow Object Detection API or PyTorch.
13. Train the models on object detection datasets and evaluate their performance.
14. Implement deep learning models for NLP tasks like sentiment analysis, named entity recognition, or machine translation using libraries like TensorFlow or PyTorch.
15. Explore pre-trained language models like BERT or Transformer for NLP tasks.

** This is only the suggested list of Practicals. Instructor may frame additional Practical's relevant to the course contents.*

Course Outcomes

- CO1: Implement and train deep learning models using TensorFlow or PyTorch.
- CO2: Apply deep learning techniques to solve real-world problems in computer vision, natural language processing, and generative modelling.
- CO3: Evaluate and analyze the performance of deep learning models on different datasets.
- CO4: Demonstrate creativity and critical thinking in designing and implementing deep learning projects.

Course Outcomes	Programme Outcomes													
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2
CO1	2	2			2								3	3
CO2		2	2	2	2								3	3
CO3				2	2				2				3	3
CO4						2		2	3	2	2	2	3	3

1 – Low, 2 – Medium, 3 - High